

TASK 3 – CAUSALITY ASSESSMENT

Topic: Evaluation of the relationship between a drug exposure and an adverse reaction using standard causality assessment methods.

1. Definition of Causality Assessment

Causality assessment is a method used for determining the causal association or relationship between drug exposure and an adverse drug reaction (ADR). It helps assess whether the suspected drug is responsible for the reported adverse event.

2. Factors Associated with Causality

Factor	Important considerations
Drug	Labeledness, relatedness, predisposing factors and concomitant medication.
Event	Whether the event is known to be drug-related or specifically related to the suspected drug.
Patient	Medical history, predisposing factors and past history.
Clinical	Temporal relationship, dechallenge/rechallenge and response to corrective treatment.

3. Major Causality Assessment Methods

A. Naranjo Adverse Drug Reaction Probability Scale

The Naranjo scale uses a structured set of questions to estimate the probability that an adverse event is related to a drug. The questions consider previous reports, temporal relationship, dechallenge, rechallenge, alternative causes, placebo response, drug concentration, dose-response relationship, previous patient experience and objective evidence.

Total score	Interpretation
≥ 9	Definite
5–8	Probable
1–4	Possible
0	Doubtful

B. WHO-UMC Causality Assessment System

Causality category	Assessment criteria
Certain	Event or laboratory test abnormality with plausible time relationship to drug intake; cannot be explained by disease or other drugs; response to withdrawal is plausible; event is definitive pharmacologically or phenomenologically; rechallenge may be satisfactory if necessary.
Probable/Likely	Event or laboratory test abnormality with reasonable time relationship to drug intake; unlikely to be attributed to disease or other drugs; response to withdrawal is clinically reasonable; rechallenge is not required.

Possible	Event or laboratory test abnormality with reasonable time relationship to drug intake; could also be explained by disease or other drugs; information on drug withdrawal may be lacking or unclear.
Unlikely	Event or laboratory test abnormality with a time relationship that makes a relationship improbable, although not impossible; disease or other drugs provide plausible explanations.
Conditional/Unclassified	Event or laboratory test abnormality; more data are needed for proper assessment or additional data are under examination.
Unassessable/Unclassifiable	Report suggests an adverse reaction, but the relationship cannot be judged because information is insufficient or contradictory and cannot be supplemented or verified.

4. Case Study: Amoxicillin and Itchy Rash

Case: On 01-Jun-2021, Raj was started on amoxicillin 500 mg three times a day for a throat infection. On 02-Jun-2021, he developed itchy rashes over his body. On 03-Jun-2021, the dose was reduced to 250 mg three times a day. On 04-Jun-2021, the rash disappeared. However, as the intensity of infection increased, the dose was increased to 500 mg three times a day on 05-Jun-2021. On 06-Jun-2021, he again developed generalized itchy rashes.

Causality Assessment Using the Naranjo Scale

Question	Assessment	Score
1. Are there previous conclusive reports on this reaction?	Yes	+1
2. Did the adverse event appear after the suspected drug was administered?	Yes	+2
3. Did the adverse reaction improve after dose reduction/discontinuation?	Yes	+1
4. Did the adverse event reappear when the drug was re-administered?	Yes	+2
5. Are there alternative causes that could independently have caused the reaction?	No	+2
6. Did the reaction reappear with placebo?	Unknown	0
7. Was the drug detected at toxic concentrations?	Unknown	0
8. Was the reaction more severe when the dose was increased or less severe when decreased?	Yes	+1
9. Did the patient have a similar reaction to the same or similar drugs previously?	Unknown	0
10. Was the adverse event confirmed by objective evidence?	Yes	+1
Total		10

Result: Total Naranjo score = 10, which falls in the **Definite** category (≥ 9). The strong temporal relationship, improvement after dose reduction, recurrence after re-exposure, and dose-response pattern support a causal association between amoxicillin and the itchy rash.

5. Adverse Event (AE) vs Adverse Drug Reaction (ADR)

Adverse Event (AE)	Adverse Drug Reaction (ADR)
Any untoward medical occurrence following use of a medicinal product, without necessarily establishing a causal relationship.	A harmful and unintended response to a medicinal product for which a causal relationship with the drug is at least a reasonable possibility.
Causality is not necessarily established.	A causal association with the drug is suspected.

Important terms shown in the training material: Serious Adverse Event (SAE) and Suspected Unexpected Serious Adverse Reactions (SUSARs).

6. Adverse Event Following Immunization (AEFI)

AEFI refers to an adverse event following immunization. The training material groups AEFI into three broad areas:

Category	Examples/description
1. Vaccine-related	Vaccine product-related and vaccine quality defect-related events.
2. Vaccination-related	Immunization error-related and immunization anxiety-related events.
3. Coincidental adverse event	An adverse event that occurs coincidentally after vaccination.

7. Five Principles Supporting Causality Assessment of Vaccine Adverse Events

The training material identifies five principles: **temporal relationship, biological plausibility, strength of association, consistency, and specificity.**

8. AEFI Causality Assessment: Main Steps

- **Step 1 – Eligibility:** Determine whether the reported event meets the criteria for assessment.
- **Step 2 – Checklist:** Collect and review relevant information about the event, immunization, timing, other causes and patient factors.
- **Step 3 – Algorithm:** Assess evidence for alternative causes, known causal association with the vaccine/vaccination, timing, evidence against causality and other qualifying factors.
- **Step 4 – Classification:** Classify the event based on the available evidence as consistent causal association, indeterminate, inconsistent causal association, or unclassifiable.

9. AEFI Classification Categories

Classification	Meaning
A. Consistent causal association to immunization	Includes vaccine product-related reaction, vaccine quality defect-related reaction, immunization error-related reaction, or immunization anxiety-related reaction.
B. Indeterminate	Evidence is insufficient or conflicting trends prevent a clear classification.
C. Inconsistent causal association to immunization	Underlying or emerging condition(s), or another cause, better explains the event than the vaccine/immunization.
Unclassifiable	Adequate information is not available; additional information is required for classification.

10. Scope for Automation

Causality assessment contains structured questions, checklists, scoring systems and classification steps. These features provide scope for automation or decision-support tools, while the final clinical/pharmacovigilance assessment should remain based on adequate evidence and expert review.

11. Key Learning Points

- Causality assessment evaluates the relationship between drug exposure and an adverse event.
- Important evidence includes temporal relationship, dechallenge, rechallenge, dose-response relationship, alternative causes and objective evidence.
- The Naranjo scale provides a numerical probability score: Definite, Probable, Possible or Doubtful.

- WHO-UMC provides clinical categories such as Certain, Probable/Likely, Possible, Unlikely, Conditional/Unclassified and Unassessable/Unclassifiable.
- AEFI assessment uses a structured eligibility, checklist, algorithm and classification process.
- Causality assessment supports consistent documentation and interpretation of suspected adverse reactions.

Source basis: Content prepared from the causality assessment training screenshots provided in this conversation, including the Naranjo scale, WHO-UMC system and AEFI causality assessment material.